Passage–2

As inflation rises, even governments previously committed to budget discipline are spending freely to help households. Higher interest rates announced by central banks are supposed to help produce modest fiscal austerity, because to maintain stable debts while paying more to borrow, governments must cut spending or raise taxes. Without the fiscal backup, monetary policy eventually loses traction. Higher interest rates become inflationary, not disinflationary, because they simply lead governments to borrow more to pay rising debt-service costs. The risk of monetary unmooring is greater when public debt rises, because interest rates become more important to budget deficits.

33. Which of the following statements best reflects/reflect the most logical and rational inference/inferences that can be made from the passage?

1. Central banks cannot bring down inflation without budgetary backing.

2. The effects of monetary policy depend on the fiscal policies pursued by the government.

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : • Inference 1 is appropriate according to the passage. If suggests that higher interest rates announced by central banks are meant to help produce fiscal austerity. Without fiscal banking monetary policy will lose its effectiveness and interest rates could be higher and situations become inflationary. So we find that budgetary backing is crucial for central banks to control inflation.

• Inference 2 is also correct according to the passage. The effects of monetary policy depends on the fiscal policies pursued by the government. This passage cleanly discusses that without the fiscal backup, monetary policy eventually loses traction and interest gets higher rate. Higher rates can lead to more borrowing by the government to pay for the rising debts – service costs. So we find that the effectiveness of monetary policy is heavily influenced by the fiscal policies of the government.

34. Based on the above passage, the following assumptions have been made:

1. Fiscal policies of governments are solely responsible for higher prices.

2. Higher prices do not affect the long-term government bonds.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Assumption 1 is not valid because according to the passage, how the government spending and fiscal policies interact with monetary policies and interest rates to impact inflation and does not suggest about fiscal policies are solely responsible for higher prices.

• **Assumption 1 is also not valid-** In this passage, we do not find this statement 'Higher prices do not affect the long term government bonds' So this passage does not discusses about the impact and interaction between long term government bond and higher prices.

35. What is the least possible number of cuts required to cut a cube into 64 identical pieces?

(a) 8 (b) 9 (c) 12 (d) 16

Ans. (b) : Number of identical pieces $64 = 4 \times 4 \times 4$
Hence, we need 3 cuts in Z direction, 3 cuts in the y-direction, and 3 cuts in the X-direction. Means our cube cut into 4 parts along Z direction and in this way 4, 4 points in x, & y directions respectively.
So minimum number of cuts required = 3 + 3 + 3 = 9.

36. In the expression 5 * 4 * 3 * 2 * 1, * is chosen from +, –, × each at most two times. What is the smallest non-negative value of the expression?

(a) 3 (b) 2 (c) 1 (d) 0

Ans. (d) : For the smallest non-negative value of the expression 5 * 4 * 3 * 2 * 1, we have to chose sign (+, –, ×) each at most two times.
So, value of expression = $5 - 4 - 3 + 2 \times 1$
$= 7 - 7 = 0$

37. A certain number of men can complete a piece of work in 6k days, where k is a natural number. By what percent should the number of men be increased so that the work can be completed in 5k days?

(a) 10% (b) (50/3)%
(c) 20% (d) 25%

Ans. (c) : Let 's assume x men can complete a price of work in 6 k days.
So, total work done, $W = x \times 6k$
Suppose y men can complete the work in 5k days.
Then we can write work done in 5k days = $y \times 5k$.
Total work is same.
Hence we can write, $y \times 5k = x \times 6k$

$$y = \frac{6}{5}k = \left(1 + \frac{1}{5}\right)x.$$

To complete the work in 5k days we have to increase the men, more than $\frac{1}{5}$ th of x, Means = $\frac{1}{5} \times 100 = 20\%$ men should be increased to complete work in 5k days.

38. X, Y and Z can complete a piece of work individually in 6 hours, 8 hours and 8 hours respectively. However, only one person at a time can work in each hour and nobody can work for two consecutive hours. All are engaged to finish the work. What is the minimum amount of time that they will take to finish the work?

(a) 6 hours 15 minutes (b) 6 hours 30 minutes
(c) 6 hours minutes (d) 7 hours

Ans. (d) : X, Y, and Z can complete a piece of work 6 hours, 8 hours and 8 hours respectively.
If total work is (LCM of 6,8,8) = 24 unit
Then x can do 4 unit in a day, y can do 3 unit in a day & Z can do 3 unit in a day.
→ However, only one person at time can work for two consecutive hours.
if x starts the work at 1st day then y at 2nd day and Z at 3rd day in this way in 3 days work can be complete
$= 4 + 3 + 3 = 10$ unit.
In next 3 day, 10 unit again will be complete then only remaining 4 unit of work will be completed by x in one days.
In this way total work can be finish in 7 (3+3+1) days.

39. How many consecutive zeros are there at the end of the integer obtained in the product $1^2 \times 2^4 \times 3^6 \times 4^8 \times ..... \times 25^{50}$?

(a) 50 (b) 55 (c) 100 (d) 200

Ans. (d) : At the end total zeros are obtained by = $(2 \times 5)^n$ where n is no of zeros.
→ $1^2 \times 2^4 \times 3^6 4^8 \times ..... 25^{50}$
To find the number of zeros at the end of the product, we need to calculated the number of 2's and number 5's or number of pair of 2 and 5.
→ In this case, number of 5's is less than the number of 2's. So we need to calculate the number of 5's.
→ There are total number of 5's $\Rightarrow 5^{10} \times 10^{20} \times 15^{30} \times 20^{40} \times 25^{50}$ only 5's $\Rightarrow 5^{10} \times 5^{20} \times 5^{30} \times 5^{100} = 5^{200}$
Total number of 5's is = 200
Hence consecutive zeros are at the end of integer is = 200

40. One January 1st, 2023, a person saved ₹ 1. On January 2nd, 2023, he saved ₹ 2 more than that on the previous day. On January 3rd, 2023, he saved ₹ 2 more than that on the previous day and so on. At the end of which date was his total savings a perfect square, as well as a perfect cube?

(a) 7th January, 2023 (b) 8th January, 2023
(c) 9th January, 2023 (d) Not possible

Ans. (d) : According to conditions which is given in the question–

Day →	1st Jan.	1nd Jan	3rd Jan	4th Jan	5th Jan	6th Jan	7th Jan	8th Jan
Saving →	1₹	3₹	5₹	7₹	9₹	11₹	13₹	15₹

Day →	9th Jan	and so on.....
Saving →	17₹	

→ There is no any number of saving which will be perfect square & cube.
→ For perfect square & cube the 64 number should be required but it is not possible that if will come in above series of days.

Directions for the following 2 (two) items:
Read the following two passages and answer the items that follow the passage. Your answers to these items should be on the passage only.

Passage–1

Unlike religion and science, poetry does not posit or expect any belief in absolute truths. Those forces or individuals who claim to have absolute truths in their grasp tend to turn dictatorial and tyrannical. Truth usually does not admit any contradictions or imperfections. It is unitarian. It is, therefore, not of much use for poetry. Poetry abides by the plurality of life and existence. Perhaps poetry follows reality which is plural, anachronistic, full of contradictions. Against the tyranny of truth, poetry remains a partisan of democratic reality. Against the arrogance of power, wealth and hierarchy, poetry proposes both humility and defiance.

41. Which one of the following statements best reflects the most logical and rational message conveyed by the above passage?

(a) It is the poetry, not science or religion, which recognizes and accepts imperfections in humans.
(b) Truth is revealed through science or religion and poetry is anathema to truth.
(c) Poetry is romantic, imaginary and is about feeling whereas science and religion are about truth.
(d) In a world of violence, tyranny and bigotry, poetry is a form of dynamic resistance.

Ans. (a) : Options (a) is correct. According to the passage, religion and science seek absolute truth while poetry relies on the plurality of life and existence, Thus, poetry recognizes and accepts the flows and contradictions inherent in human beings. Hence, the fact mentioned in options a is logical.

• Options (b) is incorrect. The passage considers science and religion as means to truth but does not discuss how poetry is anathema to truth.

• Options (c) is incorrect. The passage does not emphasis poetry as merely romantic or imaginative. It does not associate poetry with mere emotions but talks about the role of poetry in recognizing the multiplicity and contradictions of life.

• Options (d) is incorrect the passage states that the poem stands for democratic reality rather than the tyranny of truth and also expresses satisfaction that in a world of violence and fanaticism, the poem expresses both humility and defiance.

Note– Options (d) is likely to be correct. The passage mentions that "forces or individuals who claim to passage absolute truth tend to become dictators truth tend to become dictators and tyrants" and "Against the tyranny of truth, poetry stands for democratic reality. Against the arrogance of power, wealth and hierarchy, poetry proposes both humility and defiance."
Hence, it can be concluded that in a world of violence, tyranny and fanaticism, poetry is a form of dynamic resistance.

Passage–2

The flower was not invented to please us. It flaunted its petals and spread its perfume to attract an insect. The insect carries the pollen from flower to flower so that pollen is not carried away by wind and thus not wasted. What we call a flower's beauty is merely a by-product and a human invention. The perfume is not there to please us, it pleases us because it is there and we have been conditioned to it.

42. Based on the above passage, the following assumptions have been made:

1. The author of the passage believes that flowers are creations of Nature's luxury.

2. The author of the passage does not believe in the usefulness of flowers except as things of beauty.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Assumption (1) is invalid. The passage nowhere states that flowers are creations of luxury of nature. According to the passage, flowers are not created to please us, flowers complete the function of pollination by attraction insects with their fragrance. Thus flowers have a practical purpose and not to be luxurious creations.

• Assumption (2) is invalid. In the passage, the author believes that flowers have a functional utility rather than beauty. He highlights the practical role of flowers in pollination, "Insects carry pollen from one flower to another so that the wind does not blow away the pollen and thus waste it," this shows the author's recognition of the utility of flowers beyond their beauty.

43. A Question is given followed by two statement I and II. Consider the question and the statements.
Age of each of P and Q is less than 100 years but more than 10 years. If you interchange the digits of the age of P. the number represents the age of Q.
Question:
What is the difference of their ages?
Statement-I:
The age of P is greater than the age of Q
Statement-II:
The num of their ages is 11/6 times their difference.
Which one of the following is correct in respect of the above question and the statement.